

Indian Institute of Information Technology Vadodara

MS441: Discrete Mathematics

Mid semester Examination

Time: 02 : 00 hours

Total marks: 45

Name: _____

Roll No.: _____

Instructions:

1. All questions are compulsory.
2. Write down the Serial Number of the question before attempting it.
3. Solutions should be written clearly and should contain all the details.
4. Use of calculator is **NOT** allowed.
5. Each question carries 5 marks.

1. Let the universal set, $U = \{1, 2, \dots, 12\}$. Define

- $B = \{x \in U : x \text{ is divisible by } 2\}$
- $C = \{x \in U : x \text{ is divisible by } 3\}$
- $D = \{\text{primes} \leq 12\}$.

(a) Explicitly list the sets B, C, D .

(b) Compute $B \cap C, B \cup C, B \setminus D, (B \cup C) \setminus D$.

(c) Compute the symmetric difference of B and C and find its cardinality.

(d) Verify the identity $(B \cup C) \setminus D = (B \setminus D) \cup (C \setminus D)$ by direct calculation.

(e) Write down the formula for cardinality of union: $B \cup C \cup D$ and verify it for the above sets.

2. 'Prachi is creative' or 'she is intelligent.' If 'Prachi is creative,' then 'she is imaginative'. 'Prachi is not imaginative' or 'she is not a musician' In fact, 'Prachi is a musician'. Therefore 'Prachi is intelligent'. Is this argument valid? Give justification.

3. (a) Construct the truth table of $((P \vee Q) \vee R) \Leftrightarrow (P \vee (Q \vee R))$

(b) Find PCNF of $(P \Leftrightarrow Q) \vee (\neg P)$

4. Prove the following statement using a) proof by contrapositive and b) proof by contradiction.

If $3n + 2$ is divisible by 3, then n is divisible by 3.

5. Show that the set of integers, $\mathbb{Z}$ is countable. What are the cardinality of $\mathbb{Z}$ and the set of non-negative real numbers?

6. Find the GCD of 578 and 250 using the Euclidean algorithm. Also, find Bézout coefficients s and t such that

$$\text{GCD}(578, 250) = s \cdot 578 + t \cdot 250.$$

7. Find the remainder if 5^{2025} is divided by 14.

8. Explain RSA cryptosystem. Take $n = 35$. Design your own cryptosystem and demonstrate it with an example.

9. Find a bijective function between $\mathbb{Z} \times \mathbb{Z}$ and $\mathbb{Z}$ and also find the cardinality of $\mathbb{Z} \times \mathbb{Z}$.